

6.3 Let $V = \frac{\sin 3\phi}{\rho}$ in a dielectric material for which $\epsilon = 2.8 \epsilon_0$.

(a) Find $\mathbf{E}$ and $\mathbf{P}$ at point $A(1, 20^\circ, 4)$.

(b) Calculate the volume charge density at A .

6.4 Conducting sheets are located at $y = 1$ and $y = 3$ planes. The space between them is filled with a nonuniform charge distribution $\rho_v = \frac{y}{4\pi} \text{ nC/m}^3$ and $\epsilon = 4\epsilon_0$. Assuming that $V(y = 1) = 0$ and $V(y = 3) = 50 \text{ V}$, find $V(y = 2)$.

6.5 Two large flat metal sheets are located at $z = 0$ and $z = d$ and are maintained at 0 and V_0 , respectively. The charge density between the sheets is $\rho_v(z) = \rho_0 z/d$, where ρ_0 is a constant. Determine the potential at all points between the plates.

6.6 A certain material occupies the space between two conducting slabs located at $y = \pm 2 \text{ cm}$. When heated, the material emits electrons such that $\rho_v = 50(1 - y^2) \mu\text{C/m}^3$. If the slabs are both held at 30 kV, find the potential distribution within the slabs. Take $\epsilon = 3\epsilon_0$.

6.7 In cylindrical coordinates, $V = 0$ at $\rho = 2 \text{ m}$ and $V = 60 \text{ V}$ at $\rho = 5 \text{ m}$ due to charge distribution $\rho_v = \frac{10}{\rho} \text{ pC/m}^3$. If $\epsilon_r = 3.6$, find $\mathbf{E}$.

6.8 Let $U = 3xyz + y - z^2$. Show whether or not U satisfies Laplace's equation.

6.9 Given $V = x^3y + yz + cz^2$, find c such that V satisfies Laplace's equation.

6.10 In cylindrical coordinates, $V = 0$ at $\rho = 4 \text{ mm}$ and $V = V_0$ at $\rho = 12 \text{ mm}$. If $\mathbf{E} = -6\mathbf{a}_\rho \text{ kV/m}$ at $\rho = 8 \text{ mm}$, determine V_0 .

6.11 Consider the conducting plates shown in Figure 6.32. If $V(z = 0) = 0$ and $V(z = 2 \text{ mm}) = 50 \text{ V}$, determine V , $\mathbf{E}$, and $\mathbf{D}$ in the dielectric region ($\epsilon_r = 1.5$) between the plates and ρ_s on the plates.

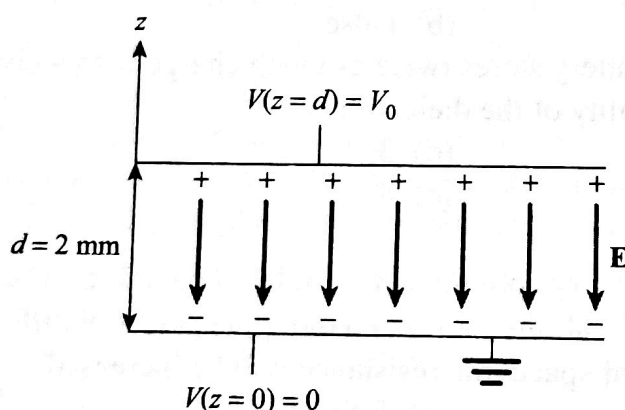


Figure 6.32 For Problem 6.11.

6.12 The cylindrical capacitor whose cross section is in Figure 6.33 has inner and outer radii of 5 mm and 15 mm, respectively. If $V(\rho = 5 \text{ mm}) = 100 \text{ V}$ and $V(\rho = 15 \text{ mm}) = 0 \text{ V}$, calculate V , $\mathbf{E}$, and $\mathbf{D}$ at $\rho = 10 \text{ mm}$ and ρ_s on each plate. Take $\epsilon_r = 2.0$.

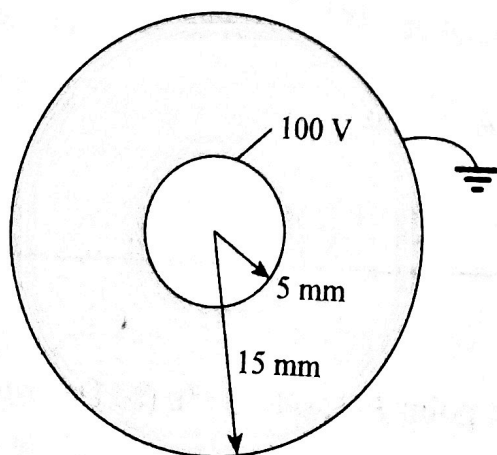


Figure 6.33 Cylindrical capacitor of Problem 6.12.

- *6.13 The potential field in a certain region is $V = (c_1 \rho^2 + c_2 \rho^{-2}) \sin 2\phi$. (a) Show that V satisfies Laplace's equation. (b) Find c_1 and c_2 if $V = 50$ V and $|\mathbf{E}| = 100$ V/m at point $P(1, 45^\circ, 1)$.
- 6.14 In cylindrical coordinates, $V = 50$ V on plane $\phi = \pi/2$ and $V = 0$ on plane $\phi = 0$. Assuming that the planes are insulated along the z -axis, determine $\mathbf{E}$ between the planes.
- *6.15 (a) Show that $V = V_0(1 - a^2/\rho^2) \rho \sin \phi$ (where V_0 is constant) satisfies Laplace's equation. (b) Determine $\mathbf{E}$ for $\rho^2 \gg a^2$.
- 6.16 Two conducting planes are located at $x = 0$ and $x = 50$ mm. The zero voltage reference is at $x = 20$ mm. Given that $\mathbf{E} = -110\mathbf{a}_x$ V/m, calculate the conductor voltages.
- 6.17 The region between concentric spherical conducting shells $r = 0.5$ m and $r = 1$ m is charge free. If $V(r = 0.5) = -50$ V and $V(r = 1) = 50$ V, determine the potential distribution and the electric field strength in the region between the shells.
- *6.18 Find V and $\mathbf{E}$ at $(3, 0, 4)$ due to the two conducting cones of infinite extent shown in Figure 6.34.

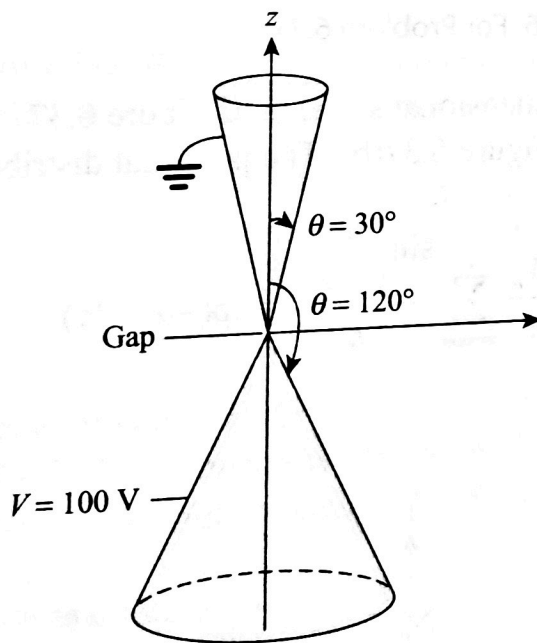


Figure 6.34 Conducting cones of Problem 6.18.

- 6.19 The inner and outer electrodes of a diode are coaxial cylinders of radii $a = 0.6$ mm and $b = 30$ mm, respectively. The inner electrode is maintained at 70 V, while the outer electrode is grounded. (a) Assuming that the length of the electrodes $\ell \gg a, b$ and ignoring the effects of space charge, calculate the potential at $\rho = 15$ mm. (b) If an electron is injected radially through a small hole in the inner electrode with velocity 10^7 m/s, find its velocity at $\rho = 15$ mm.
- 6.20 An electrode with a hyperbolic shape ($xy = 4$) is placed above a grounded right-angle corner as in Figure 6.35. Calculate V and $\mathbf{E}$ at point $(1, 2, 0)$ when the electrode is connected to a 20 V source.

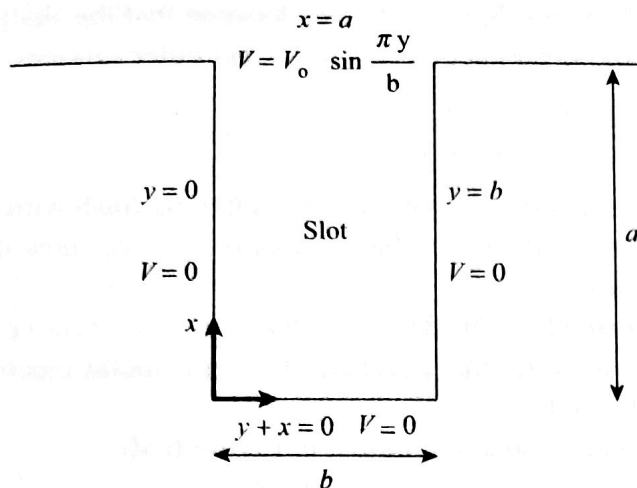


Figure 6.38 For Problem 6.24.

- 6.25 By letting $V(\rho, \phi) = R(\rho)\Phi(\phi)$ be the solution of Laplace's equation in a region where $\rho \neq 0$, show that the separated differential equations for R and Φ are

$$R'' + \frac{R'}{\rho} - \frac{\lambda}{\rho^2} R = 0$$

and

$$\Phi'' + \lambda \Phi = 0$$

where λ is the separation constant.

- 6.26 A potential in spherical coordinates is a function of r and θ but not ϕ . Assuming that $V(r, \theta) = R(r)F(\theta)$, obtain the separated differential equations for R and F in a region which $\rho_v = 0$.

Section 6.5—Resistance and Capacitance

- 6.27 Show that the resistance of the bar of Figure 6.17 between the vertical ends located at $\phi = 0$ and $\phi = \pi/2$ is

$$R = \frac{\pi}{2\sigma t \ln \frac{b}{a}}$$

- *6.28 Show that the resistance of the sector of a spherical shell of conductivity σ , with cross section shown in Figure 6.39 (where $0 \leq \phi < 2\pi$), between its base (i.e., from $r = a$ to $r = b$) is

$$R = \frac{1}{2\pi\sigma(1 - \cos\alpha)} \left[\frac{1}{a} - \frac{1}{b} \right]$$

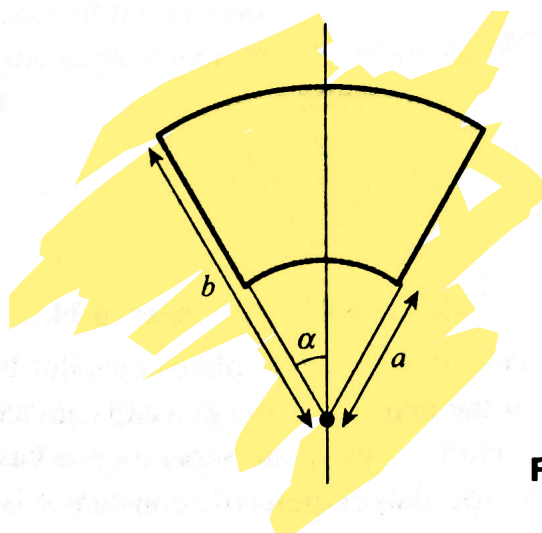


Figure 6.39 For Problem 6.28.

- **6.29** A spherical shell has inner and outer radii a and b , respectively. Assume that the shell has a uniform conductivity σ and that it has copper electrodes plated on the inner and outer surfaces. Show that

$$R = \frac{1}{4\pi\sigma} \left(\frac{1}{a} - \frac{1}{b} \right)$$

- ***6.30** A hollow conducting hemisphere of radius a is buried with its flat face lying flush with the earth's surface, thereby serving as an earthing electrode. If the conductivity of earth is σ , show that the leakage conductance between the electrode and earth is $2\pi a\sigma$.

- 6.31** In an integrated circuit, a capacitor is formed by growing a silicon dioxide layer ($\epsilon_r = 4$) of thickness $1\text{ }\mu\text{m}$ over the conducting silicon substrate and covering it with a metal electrode of area S . Determine S if a capacitance of 2 nF is desired.

- **6.32** Calculate the capacitance of the parallel-plate capacitor shown in Figure 6.40.

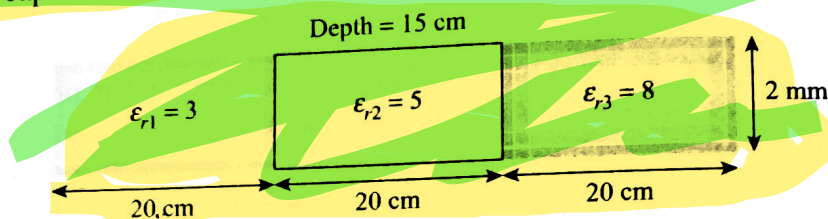


Figure 6.40 For Problem 6.32.

- **6.33** Evaluate the capacitance of the parallel-plate capacitor with multilayer dielectric shown in Figure 6.41.

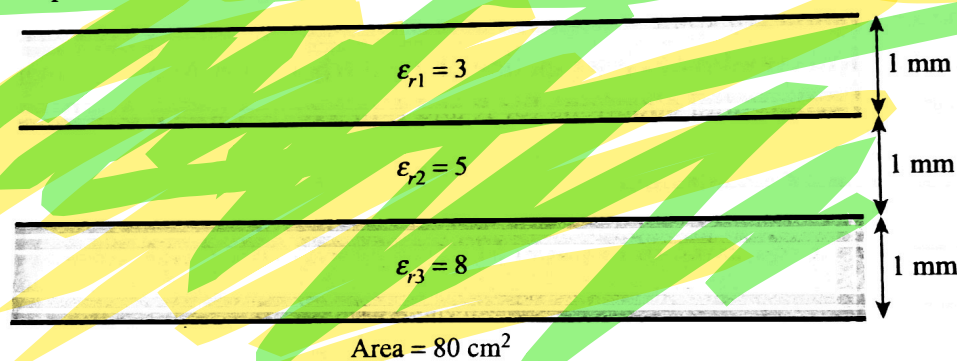


Figure 6.41 For Problem 6.33.

- 6.34** The parallel-plate capacitor of Figure 6.42 is quarter-filled with mica ($\epsilon_r = 6$). Find the capacitance of the capacitor.

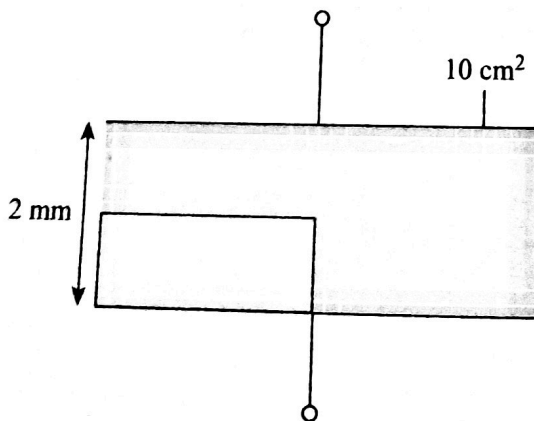


Figure 6.42 For Problem 6.34.

- 6.35** To appreciate the physical size of 1 F capacitor, consider a parallel-plate capacitor filled with air and with separation distance of 1 mm . Find the area of the plates to provide a capacitance of 1 F .
- ***6.36** An air-filled parallel plate capacitor of length L , width a , and plate separation d has its plates maintained at constant potential difference V_0 . If a dielectric slab of dielectric constant ϵ_r is slid between the

plates and is withdrawn until only a length x remains between the plates as in Figure 6.43, show that the force tending to restore the slab to its original position is

$$F = \frac{\epsilon_0(\epsilon_r - 1) a V_0^2}{2d}$$

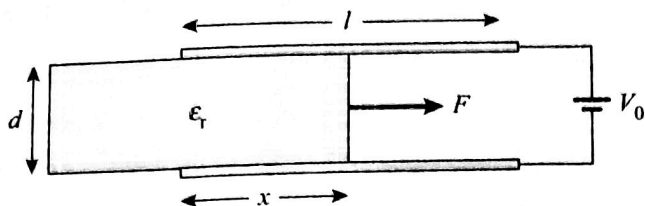
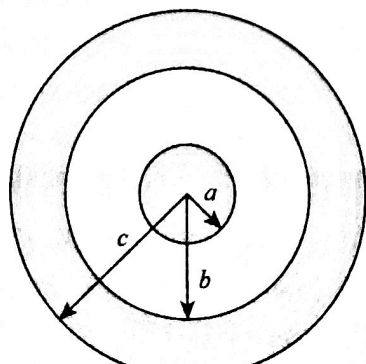


Figure 6.43 For Problem 6.36.

- 6.37 The space between spherical conducting shells $r = 5$ cm and $r = 10$ cm is filled with a dielectric material for which $\epsilon = 2.25\epsilon_0$. The two shells are maintained at a potential difference of 80 V. (a) Find the capacitance of the system. (b) Calculate the charge density on shell $r = 5$ cm.
- 6.38 The capacitance of a parallel-plate capacitor is $56 \mu\text{F}$ when the dielectric material is in place. The capacitance drops to $32 \mu\text{F}$ when the dielectric material is removed. Calculate the dielectric constant ϵ_r of the material.
- 6.39 A large parallel-plate capacitor has one plate at $z = 0$ and maintained at 40 V, while the other plate at $z = 2$ mm is maintained at 0 V. Find the potential between the plates.
- 6.40 A parallel-plate capacitor has a 4 mm plate separation, 0.5 m^2 surface area per plate, and a dielectric with $\epsilon_r = 6.8$. If the plates are maintained at 9 V potential difference, calculate (a) the capacitance, (b) the charge density on each plate.
- 6.41 A parallel-plate capacitor with plate area S and spacing d has a charge Q on each plate. Assuming that the space between the plates is filled with dielectric $\epsilon = \epsilon_r \epsilon_0$, determine the energy stored when the plate spacing is (a) doubled, (b) halved.
- 6.42 A parallel-plate capacitor has separation 5 mm and area 0.4 m^2 . If the space between the plates is filled with dielectric with $\epsilon_{r1} = 2.5$, $0 < d < 1.5$ mm, dielectric with $\epsilon_{r2} = 5.6$, $1.5 \text{ mm} < d < 3$ mm, and dielectric with $\epsilon_{r3} = 8.1$, $3 \text{ mm} < d < 5$ mm, calculate the capacitance.
- 6.43 A parallel-plate capacitor has plate area 200 cm^2 and plate separation of 3 mm. The charge density is $1 \mu\text{C}/\text{m}^2$ and air is the dielectric. Find
- The capacitance of the capacitor
 - The voltage between the plates
 - The force with which the plates attract each other
- 6.44 A spherical capacitor has inner radius d and outer radius a . Concentric with the spherical conductors and lying between them is a spherical shell of outer radius c and inner radius b . If the regions $d < r < c$, $c < r < b$, and $b < r < a$ are filled with materials with permittivities ϵ_1 , ϵ_2 , and ϵ_3 , respectively, determine the capacitance of the system.
- 6.45 Determine the capacitance of a conducting sphere surrounded by a thick spherical shell as shown in Figure 6.44.



- 6.46** A coaxial cable has inner radius of 5 mm and outer radius of 8 mm. If the cable is 3 km long, calculate its capacitance. Assume $\epsilon = 2.5\epsilon_0$.
- 6.47** A capacitor consists of two plates with equal width $(b - a)$, and a length L in the z -direction. The plates are separated by $\phi = \pi/4$, as shown in Figure 6.45. Assume that the plates are separated by a dielectric material ($\epsilon = \epsilon_0\epsilon_r$) and ignore fringing. Determine the capacitance.

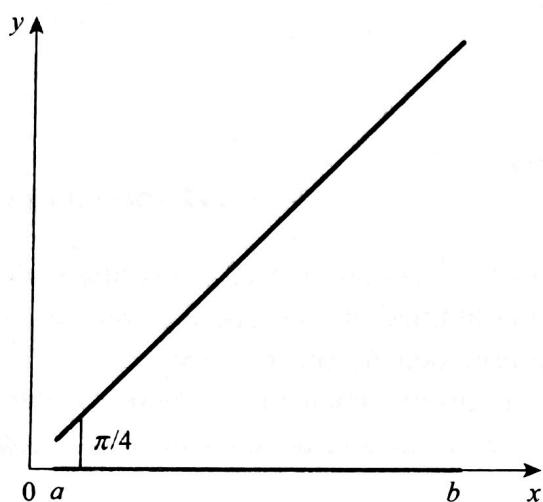


Figure 6.45 For Problem 6.47.

- 6.48** Calculate the capacitance of a coaxial cable with inner radius 1 mm, outer radius 3.5 mm, and length 400 mm. Assume the dielectric has $\epsilon = 4.2\epsilon_0$.
- *6.49** In an ink-jet printer the drops are charged by surrounding the jet of radius $20\ \mu\text{m}$ with a concentric cylinder of radius $600\ \mu\text{m}$ as in Figure 6.46. Calculate the minimum voltage required to generate a charge 50 fC on the drop if the length of the jet inside the cylinder is $100\ \mu\text{m}$. Take $\epsilon = \epsilon_0$.

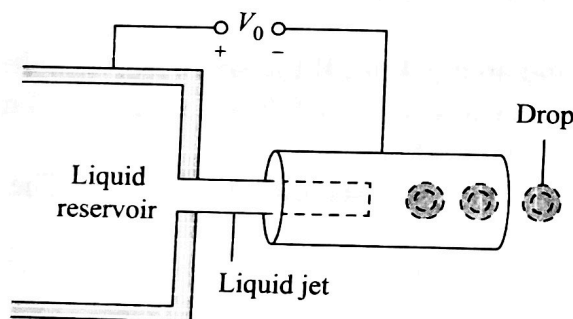


Figure 6.46 Simplified geometry of an ink-jet printer; for Problem 6.49.

- 6.50** The cross section of a cable is shown in Figure 6.47. Determine the capacitance per unit length.

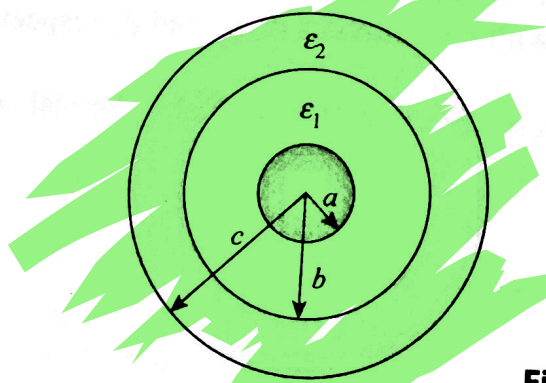


Figure 6.47 For Problem 6.50.

- *6.51** A spherical capacitor has an inner conductor of radius a carrying charge Q and is maintained at zero potential. If the outer conductor contracts from a radius b to c under internal forces, prove that the work performed by the electric field as a result of the contraction is

$$W = \frac{Q^2(b-c)}{8\pi\epsilon bc}$$

- *6.52 A parallel-plate capacitor has its plates at $x = 0, d$ and the space between the plates is filled with an inhomogeneous material with permittivity $\epsilon = \epsilon_0 \left(1 + \frac{x}{d}\right)$. If the plate at $x = d$ is maintained at V_0 while the plate at $x = 0$ is grounded, find:

- (a) V and $\mathbf{E}$
- (b) $\mathbf{P}$
- (c) ρ_{ps} at $x = 0, d$
- (d) the capacitance, assuming that each plate has area S

- 6.53 Two parallel conducting plates are located at $x = d$ and $x = -d$. The plate at $x = d$ is held at V_0 , while the plate at $x = -d$ is grounded. If the space between the plates is filled with an inhomogeneous dielectric medium with

$$\epsilon = \frac{2\epsilon_0}{1 + \left(\frac{x}{d}\right)^2}$$

find the capacitance. Assume that each plate has an area S .

- 6.54 A spherical capacitor has inner radius a and outer radius b and is filled with an inhomogeneous dielectric with $\epsilon = \epsilon_0 k/r^2$. Show that the capacitance of the capacitor is

$$C = \frac{4\pi\epsilon_0 k}{b-a}$$

- 6.55 If the earth is regarded a spherical capacitor, what is its capacitance? Assume the radius of the earth to be approximately 6370 km.
- 6.56 A capacitor is formed by two coaxial metal cylinders of radii $a = 1$ mm and $b = 5$ mm. If the space between the cylinders is filled with a dielectric having $\epsilon_r = 3(1 + \rho)$, $a < \rho < b$, and ρ is in millimeters, determine the capacitance per meter.

Section 6.6—Method of Images

- 6.57 In free space, infinite planes $y = 4$ and $y = 8$ carry charges 20 nC/m^2 and 30 nC/m^2 , respectively. If plane $y = 2$ is grounded, calculate $\mathbf{E}$ at $P(0, 0, 0)$ and $Q(-4, 6, 2)$.
- 6.58 Two point charges of 3 nC and -4 nC are placed, respectively, at $(0, 0, 1 \text{ m})$ and $(0, 0, 2 \text{ m})$ while an infinite conducting plane is at $z = 0$. Determine
- (a) The total charge induced on the plane
 - (b) The magnitude of the force of attraction between the charges and the plane
- *6.59 A point charge of $10 \mu\text{C}$ is located at $(1, 1, 1)$, and the positive portions of the coordinate planes are occupied by three mutually perpendicular plane conductors maintained at zero potential. Find the force on the charge due to the conductors.
- 6.60 A point charge Q is placed between two earthed intersecting conducting planes that are inclined at 45° to each other. Determine the number of image charges and their locations.
- 6.61 Infinite line $x = 3, z = 4$ carries 16 nC/m and is located in free space above the conducting plane $z = 0$.
- (a) Find $\mathbf{E}$ at $(2, -2, 3)$.
 - (b) Calculate the induced surface charge density on the conducting plane at $(5, -6, 0)$.
- *6.62 A grounded metal sheet is located in the $z = 0$ plane, while a point charge Q is located at $(0, 0, a)$. Find the force acting on a point charge $-Q$ placed at $(a, 0, a)$.